AIM

To write a C program to perform bitwise operations such as Set Bit, Reset Bit, Read Bit, and Toggle Bit using functions, and to perform string operations like reversing a string and converting an integer to a string using user-defined functions.

PROCEDURE

1. Include the required header file stdio.h.
2. Declare function SetBit() ResetBit() ReadBit() ToggleBit()
3. Declare function ReverseString() IntToString()
4. In the main() function:
   - Get an integer number and bit position from the user.
   - Call each bitwise function and display the results.
   - Get a string from the user and reverse it using ReverseString().
   - Get an integer from the user and convert it into a string using IntToString().
5. Compile and execute the program.

CODE

```
/*

Name of the File   : function.c

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Purpose            :To perform bitwise operations (SetBit, ResetBit, ReadBit, ToggleBit) and string operation (Reverse String and Integer to String conversion) using functions in C.

*/

#include <stdio.h>

int SetBit(int Num, int Pos);

int ResetBit(int Num, int Pos);

int ReadBit(int Num, int Pos);

int ToggleBit(int Num, int Pos);

void ReverseString(char Str[]);

void IntToString(int Num, char Str[]);

int main()

{

   int Number, Position, InputNum;
```

```c
    char String[50];
    char ResultStr[50];
    printf("Enter a number: ");
    scanf("%d", &Number);
    printf("Enter bit position: ");
    scanf("%d", &Position);
    printf("\nOriginal Number: %d\n", Number);
    printf("After SetBit: %d\n", SetBit(Number, Position));
    printf("After ResetBit: %d\n", ResetBit(Number, Position));
    printf("ReadBit: %d\n", ReadBit(Number, Position));
    printf("After ToggleBit: %d\n\n", ToggleBit(Number, Position));
    printf("Enter a string: ");
    scanf(" %[^\n]", String);
    ReverseString(String);
    printf("Reversed String: %s\n\n", String);
    printf("Enter an integer to convert into string: ");
    scanf("%d", &InputNum);
    IntToString(InputNum, ResultStr);
    printf("Converted String: %s\n", ResultStr);
    return 0;
}
int SetBit(int Num, int Pos)
{return Num | (1 << Pos);}
int ResetBit(int Num, int Pos)
{return Num & ~(1 << Pos);}
int ReadBit(int Num, int Pos)
{return (Num >> Pos) & 1;}
int ToggleBit(int Num, int Pos)
{return Num ^ (1 << Pos);}
void ReverseString(char Str[])
{
    int Start = 0, End = 0;
```

```
    char Temp;
    while (Str[End] != '\0')
        End++;
    End--;
    while (Start < End)
    {
        Temp = Str[Start];
        Str[Start] = Str[End];
        Str[End] = Temp;
        Start++;
        End--;
    }
}
void IntToString(int Num, char Str[])
{
    int Index = 0, IsNegative = 0;
    if (Num == 0)
    {
        Str[Index++] = '0';
        Str[Index] = '\0';
        return;
    }
    if (Num < 0)
    {
        IsNegative = 1;
        Num = -Num;
    }
    while (Num > 0)
    {
        Str[Index++] = (Num % 10) + '0';
        Num /= 10;
    }
```

```
    if (IsNegative)
        Str[Index++] = '-';
    Str[Index] = '\0';
    ReverseString(Str);
}
```

OUTPUT

```
PS C:\AMI C PROGRAM> cd "c:\AMI C PROGRAM\Monish_kanna_week2_practice2\" ; if ($?)
{ gcc function.c -o function } ; if ($?) { .\function }
Enter a number: 10
Enter bit position:
2

Original Number: 10
After SetBit: 14
After ResetBit: 10
ReadBit: 0
After ToggleBit: 14

Enter a string: American MegaTrends
Reversed String: sdnerTageM naciremA

Enter an integer to convert into string: 2026
Converted String: 2026
```

RESULT

Thus, the C program to perform bitwise operations and string operations using functions was successfully executed, and the output was obtained as expected.